

# Confronting Misinformation - Detecting and Verifying Fake News on Social Media

## Objective

The goal is to build an AI-powered system that:

- Accepts multiple types of inputs: text, news links (URLs), images, and videos.
- Predicts whether the news is Real or Fake using machine learning and NLP.
- Verifies the claim by searching credible online sources.
- Generates a "Community Notes"-style contrast report highlighting confirmed facts and contradictions.

## Why Streamlit?

We are using Streamlit because:

- It allows rapid prototyping entirely in Python.
- It easily supports text, file uploads (image/video), and URL inputs.
- It can display results with rich formatting, colors, and media previews.
- It can be deployed for free using Streamlit Cloud.

## Input Types

The system supports four input types:

1. Text - user enters a news headline or paragraph.
2. URL - user pastes a news article link.
3. Image - user uploads a photo (screenshot of news).
4. Video - user uploads a short video (news clip).

## Processing Pipeline

1. Input Handling:

- Text: Directly sent to model.
- URL: Extract content using newspaper3k / BeautifulSoup.
- Image: OCR extraction with pytesseract; optional reverse image search.
- Video: Audio transcription via Whisper / Google STT.

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## 2. Fake News Prediction:

- Preprocessing: Stopword removal, punctuation removal, lowercasing, tokenization, lemmatization.
- Feature Extraction: TF-IDF or Transformer embeddings (BERT, RoBERTa).
- Model: Logistic Regression / SVM baseline, upgrade to fine-tuned BERT.

## 3. Online Verification:

- Use Google/Bing Search APIs to fetch top 5 credible articles from trusted domains.
- Extract text for fact comparison.

## 4. Fact Matching & Contrast:

- Semantic similarity (Sentence-BERT).
- Highlight confirmed and contradicted facts.
- Generate "Community Notes"-style summary with references.

## Example Output

Prediction: Fake (92% confidence)

Sources Found:

- BBC News - "NASA denies alien sightings"
- NASA Blog - "Astronomy update for August"

Community Note:

The claim is contradicted by official NASA statements and BBC News. No credible outlet has confirmed the alleged event.

## Techniques & Technologies

- Text preprocessing: NLTK, spaCy
- Image to text: OCR (pytesseract), OpenCV
- Video to text: Whisper, Google STT
- Feature extraction: TF-IDF, Word2Vec, BERT embeddings
- Prediction model: Logistic Regression, SVM, BERT
- Web verification: Google/Bing Search API

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- Fact matching: Sentence-BERT (cosine similarity)
- UI: Streamlit
- Deployment: Streamlit Cloud / Heroku / AWS

## Final Flow in Streamlit

1. User selects input type (Text, URL, Image, Video).
2. System extracts/converts input to a text claim.
3. Model predicts Fake or Real.
4. Web verification fetches credible sources.
5. Fact matching generates contrast report.
6. Streamlit displays prediction, confidence score, credible sources, and Community Notes summary.